

TLD file, the cell and the location at which the TLD file is to be saved.

Although it is essential to generate TLD for identifying hierarchy, LASI needs TLC files for editing, where a TLC file gets updated every time Save is pressed, leaving behind the TLD file unchanged.

Now, if we want to edit a previously-created diagram, we need to use Edit-in-Place (EIP) mode, where modification will take place based on hierarchy. It works with the use of Cget command, after which Ctrl key on the keyboard and Load on the software needs to be pressed. This allows cells in the lower ranking to be modified with respect to objects from the cells in the higher ranking.

Utilities add to LASI

LASI has several utility programs like crystallographic information file (CIF), drawing exchange format (DXF) and GDS format converters, along with a user programmable bitmap based design rule check (DRC), Spice netlist compiler and a basic matrix router. Spice netlist compiler in LASI extracts Spice circuit files from both the layout and the schematic but never simulates these. So, at times LASI can be considered a compiler but never a simulator.

DRC, on the other hand, proves to be an added advantage for LASI. It becomes quite hectic to design a layout with one tool and then use another tool to check it. LASI has LasiDrc for this, which runs DRC on the file name thus selected and then identifies errors, if any. Pressing esc on the keyboard can abort this program anytime. The only point to make note of is that the correct file name should be associated with DRC.

Some more features

LASI was originally designed for the complementary metal oxide semiconductor (CMOS) logic circuit layout but as discussed before, the current version of this tool is not limited to CMOS and is versatile. Other than versatility, LASI has a few more features that can be counted as benefits.



Fig. 2: Layout editing window

Installation instructions

1. Unzip software to get the executable file.
2. Run the executable and follow instructions that come up. It will either replace or build files in the main Lasi7 folder and sub-folders.
3. If there are any other LASI drawings previously downloaded, these should not get affected unless there are special mentions in the installation.
4. If you have downloaded any add-on files from circuits and resources page, then use the same procedure mentioned above to install these. Running the executable file in this case will save all drawings in their respective sub-folders.
5. If there are any zip files that do not have any executables for self-installing, then follow the steps given below:
 - Copy and paste the zip file in \Lasi7\Zips sub-directory.
 - Run the main installation executable again.

System programs will not only get reinstalled but will also let you choose from drawings from zips sub-folder that you want to install.

Note: The sub-folder will have the same name as the zip file.

Usually, we do not have any choice over OSes for CAD tools but a 64-bit Windows OS is always considered the best choice because of its large addressing capability. On top of that, using CAD tools for educational purposes is quite difficult. LASI proves that it is not the same for all CAD tools. If you are working with LASI, you do not need to be an expert on technology, but you just need to consider IC design as art.

LASI allows constant backup of work in a different location other than the hard disk being used by this tool. This helps save space on the hard disk being used for the tool. However, LASI does not have any backup button for the same. This can be done by exporting TLD files to that different location.

Users' reviews

Users' reviews account for a lot when

finding out about the importance, usage and working experience of a product. This applies to software tools as well, especially when these are design tools and available free of cost. Also, selection of the correct design tool at the correct price is quite difficult these days.

A user at www.researchgate.net/ suggests LASI to another user for designing a lithography mask in micron scale because of its ease of use and the large number of tutorials available for support. Whereas another user, who appreciates LASI as freeware, suggests that it requires more learning time than other layout editors.

However, a user on www.memsmnet.org informs he has used LASI exclusively when he was still in college and claims it to be a great program. ●

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